

# BASICS OF CIVIL AND MECHANICAL ENGINEERING

## PART-A

### UNIT-1

1. Explain the role of Mechanical Engineering in Industries and also for the society.
2. Explain about the (a) Energy Sector (b) Automotive Sector (c) Aerospace Sector (d) Marine Sector.
3. Explain about 4 types of Ferrous materials and non-ferrous materials in detail.
4. Define Ceramics and explain about the classification, properties of ceramics along with advantages and disadvantages of ceramics.
5. Define Composite materials and explain about the classification of composite materials along with advantages and disadvantages of composite materials.

### UNIT -2

1. Explain about the terms used in Casting with neat sketch.
2. Explain about (a) Hot Working (b) Cold Working with advantages and disadvantages.
3. Write the types of Welding. Explain (a) Gas welding (b) Oxy-Acetylene Welding with neat sketch.
4. Explain the working of (a)Turning Process (b) Milling Process with neat sketch.
5. Briefly explain about the working principle, functions, classification of Boiler.
6. Derive an expression for efficiency of Otto cycle by explaining all the processes in it, along with P-V and T-S diagrams.
7. Derive an expression for efficiency of Diesel cycle by explaining all the processes in it, along with P-V and T-S diagrams.
8. Explain the working Principle of 4-Stroke SI engine with neat sketch.
9. What are the different components of an Electric Vehicle. Explain them in detail.

### UNIT-3

1. Explain the Principle of Operation of Steam Power Plant with a neat sketch.
2. Explain the Working of Nuclear Power Plant with neat sketch along with advantages and disadvantages.
3. Explain about (a) Belt drives and its types with neat sketch (b) chain drives and its types with neat sketch.
4. Explain about the components of a robot and write short notes on Links.
5. Write about the 2 different types of Translation motion with neat sketch and also explain 3 types of Rotary Motion with neat sketch.
6. Explain about the 5 types of Configurations of robot with advantages and disadvantages.